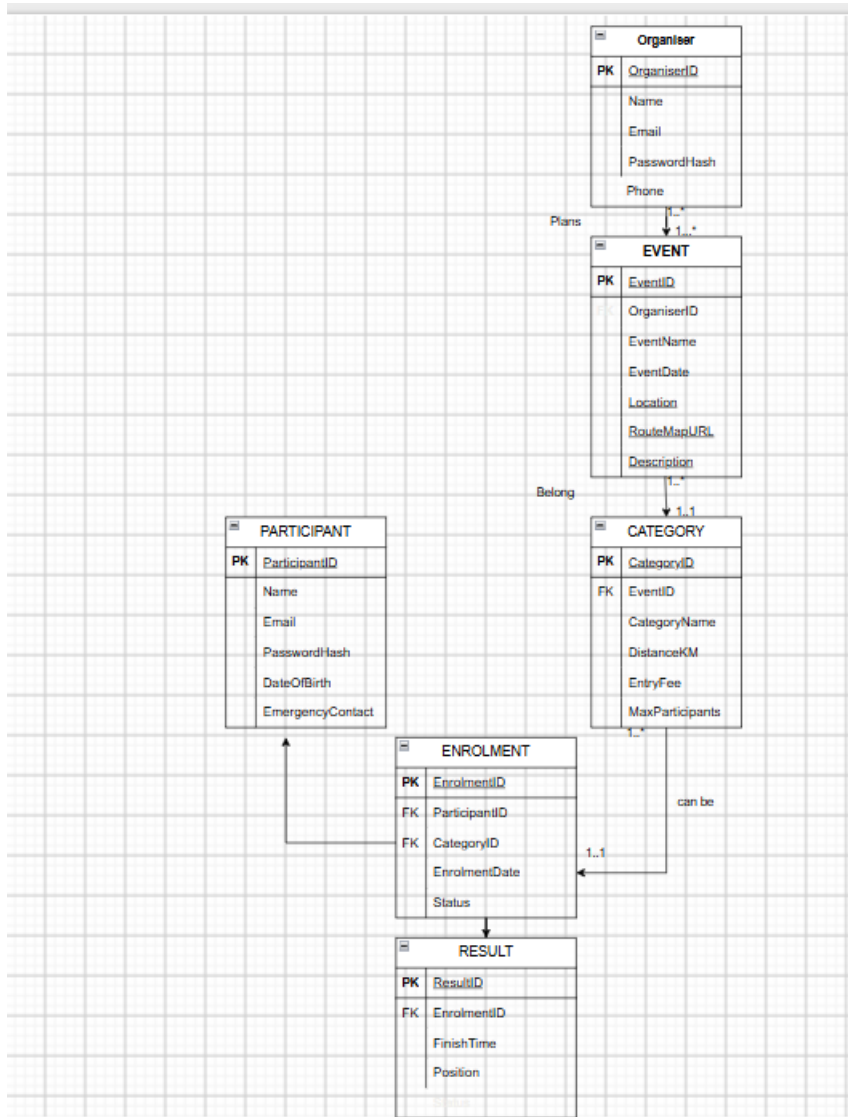


SECTION A



SECTION B

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as either an Organiser or Participant	None	{ name, email, password, role }	201 Created – user object (no password) 400 Bad Request – validation failed 409 Conflict – email already in use
POST	/api/auth/login	Authenticates a user and returns a JWT	None	{ email, password }	200 OK – token + user info 401 Unauthorized – invalid credentials
GET	/api/users/me	Returns the logged-in user's own profile	Any	None	200 OK – profile object 401 Unauthorized
PUT	/api/users/me	Updates the logged-in user's own profile	Any	{ name, phone, ... }	200 OK – updated profile 400 Bad Request
GET	/api/events	Lists all upcoming events (public browsing)	None	None	200 OK – array of events
GET	/api/events/{id}	Gets full detail for one event	None	None	200 OK – event

					object 404 Not Found
POST	/api/events	Creates a new event	Organiser	{ eventName, eventDate, location, description, routeMapUrl }	201 Created – event object 400 Bad Request 403 Forbidden
PUT	/api/events/{id}	Updates an event	Organiser (owner)	{ eventName, eventDate, location, description, routeMapUrl }	200 OK – updated event 403 Forbidden 404 Not Found
DELETE	/api/events/{id}	Deletes an event	Organiser (owner)	None	204 No Content 403 Forbidden 404 Not Found
GET	/api/events/{eventId}/categories	Lists all categories for a specific event	None	None	200 OK – array of categories 404 Not Found
POST	/api/events/{eventId}/categories	Adds a new category to an event	Organiser (owner)	{ categoryName, distanceKm, entryFee, maxParticipants }	201 Created – category object 403 Forbidden 404 Not Found
PUT	/api/categories/{id}	Updates a category	Organiser (owner)	{ categoryName, distanceKm	200 OK – updated category 403

				, entryFee, maxParticipants }	Forbidden 404 Not Found
DELETE	/api/categories/{id}	Deletes a category	Organiser (owner)	None	204 No Content 403 Forbidden 404 Not Found
POST	/api/enrolments	Enrols the logged-in participant into a category	Participant	{ categoryId }	201 Created – enrolment object 400 Bad Request 409 Conflict – already enrolled
GET	/api/enrolments/me	Lists the logged-in participant's own enrolments	Participant	None	200 OK – array of enrolments
GET	/api/events/{eventId}/enrolments	Lists all participants enrolled in an event	Organiser (owner)	None	200 OK – array of enrolments 403 Forbidden
DELETE	/api/enrolments/{id}	Cancels the logged-in participant's own enrolment	Participant (owner)	None	204 No Content 403 Forbidden 404 Not Found
POST	/api/enrolments/{id}/result	Captures a race result for a participant's enrolment	Organiser (owner)	{ finishTime, position, status }	201 Created – result object 400 Bad Request 403 Forbidden

					n 404 Not Found
PUT	/api/results/{id}	Updates an existing result	Organiser (owner)	{ finishTime, position, status }	200 OK – updated result 403 Forbidden 404 Not Found
GET	/api/results/me	Shows the logged-in participant's personal results history	Participant	None	200 OK – array of results
GET	/api/events/{eventId}/results	Public leaderboard/results for an event	None	None	200 OK – array of results 404 Not Found

SECTION C

```
--CREATE DATABASE RaceDayDB;
```

```
USE RaceDayDB;
```

```
CREATE TABLE Roles (
  RoleID INT IDENTITY(1,1) PRIMARY KEY,
  RoleName VARCHAR(20) NOT NULL UNIQUE
);
```

```
CREATE TABLE Users (
  UserID INT IDENTITY(1,1) PRIMARY KEY,
  FullName VARCHAR(100) NOT NULL,
  Email VARCHAR(150) NOT NULL UNIQUE,
```

```
PasswordHash VARCHAR(255) NOT NULL,  
RoleID INT NOT NULL,  
CreatedAt DATETIME NOT NULL DEFAULT GETDATE(),  
FOREIGN KEY (RoleID) REFERENCES Roles(RoleID)  
);
```

```
CREATE TABLE Events (  
EventID INT IDENTITY(1,1) PRIMARY KEY,  
EventName VARCHAR(150) NOT NULL,  
EventDate DATE NOT NULL,  
Location VARCHAR(150) NOT NULL,  
Description VARCHAR(500) NULL,  
OrganiserID INT NOT NULL,  
CreatedAt DATETIME NOT NULL DEFAULT GETDATE(),  
FOREIGN KEY (OrganiserID) REFERENCES Users(UserID)  
);
```

```
CREATE TABLE Categories (  
CategoryID INT IDENTITY(1,1) PRIMARY KEY,  
EventID INT NOT NULL,  
CategoryName VARCHAR(100) NOT NULL,  
DistanceKm DECIMAL(5,2) NOT NULL,  
EntryFee DECIMAL(8,2) NOT NULL DEFAULT 0,  
MaxParticipants INT NOT NULL DEFAULT 100,  
FOREIGN KEY (EventID) REFERENCES Events(EventID)  
);
```

```
CREATE TABLE EventEnrolments (  

```

```
EnrolmentID INT IDENTITY(1,1) PRIMARY KEY,  
ParticipantID INT NOT NULL,  
CategoryID INT NOT NULL,  
EnrolmentDate DATETIME NOT NULL DEFAULT GETDATE(),  
Status VARCHAR(20) NOT NULL DEFAULT 'Confirmed',  
FOREIGN KEY (ParticipantID) REFERENCES Users(UserID),  
FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID),  
UNIQUE (ParticipantID, CategoryID)  
);
```

```
CREATE TABLE Results (  
ResultID INT IDENTITY(1,1) PRIMARY KEY,  
EnrolmentID INT NOT NULL UNIQUE,  
FinishTime TIME NULL,  
Position INT NULL,  
CapturedAt DATETIME NOT NULL DEFAULT GETDATE(),  
FOREIGN KEY (EnrolmentID) REFERENCES EventEnrolments(EnrolmentID)  
);
```

```
INSERT INTO Roles (RoleName) VALUES ('Organiser'), ('Participant');
```

```
INSERT INTO Users (FullName, Email, PasswordHash, RoleID)  
VALUES  
( 'Thabo Nkosi', 'thabo.nkosi@raceday.co.za', 'HASHED_PASSWORD_1', 1),  
( 'Lindiwe Dube', 'lindiwe.dube@raceday.co.za', 'HASHED_PASSWORD_2', 1),  
( 'Sarah Botha', 'sarah.botha@example.com', 'HASHED_PASSWORD_3', 2),  
( 'Kagiso Molefe', 'kagiso.molefe@example.com', 'HASHED_PASSWORD_4', 2);
```

INSERT INTO Events (EventName, EventDate, Location, Description, OrganiserID)

VALUES

('Cape Town Cycle Tour', '2026-03-08', 'Cape Town', 'Iconic 109km cycling road race around the peninsula.', 1),

('Soweto Marathon', '2026-11-01', 'Soweto', 'Community road running event through historic Soweto.', 1),

('Two Oceans Marathon', '2026-04-04', 'Cape Town', 'Ultra marathon along the Cape Peninsula coastline.', 2);

INSERT INTO Categories (EventID, CategoryName, DistanceKm, EntryFee, MaxParticipants)

VALUES

(1, 'Individual Time Trial', 109.00, 550.00, 500),

(1, 'Mini Cycle Tour', 35.00, 250.00, 300),

(2, '10km Fun Run', 10.00, 100.00, 1000),

(2, '21km Half Marathon', 21.10, 200.00, 800),

(3, 'Ultra Marathon', 56.00, 650.00, 400),

(3, 'Half Marathon', 21.10, 300.00, 600);

INSERT INTO EventEnrolments (ParticipantID, CategoryID, Status)

VALUES

(3, 1, 'Confirmed'),

(3, 4, 'Confirmed'),

(4, 3, 'Confirmed'),

(4, 5, 'Pending');

INSERT INTO Results (EnrolmentID, FinishTime, Position)

VALUES

(1, '03:12:45', 214),

(2, '01:45:10', 58),

(3, '00:48:22', 12);